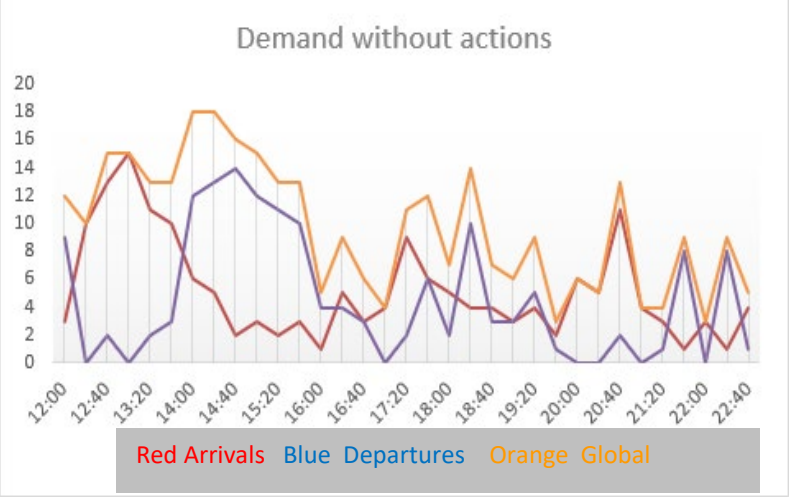
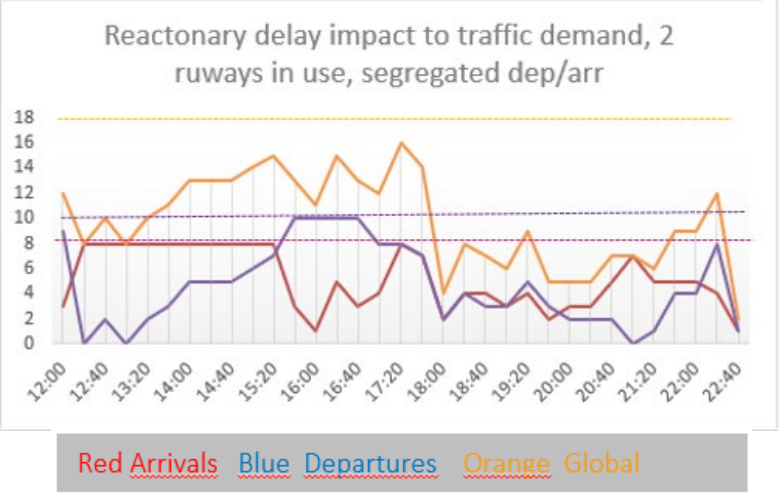
Balancing of arrival and departure flows at an airport with airports with pronounced arrival and departure peaks. Helsinki case study

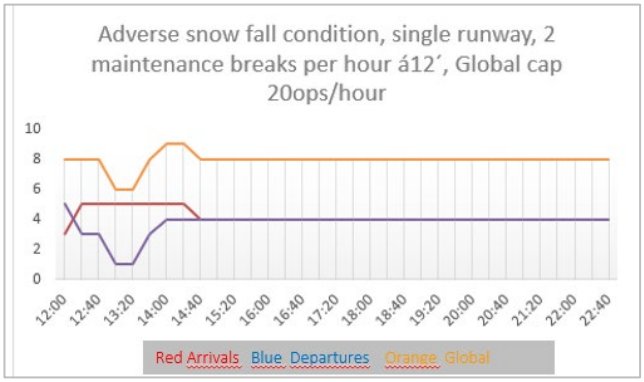
Helsinki is a typical hub-airline airport, where Finnair is not big enough to fill up all operating hours and the traffic pattern is structured in waves-early morning departure wave, followed by an arrival mid-day peak form Southern and Central Europe fetching passengers to the Far East destinations forming a departure wave in the early afternoon. Finally, a smaller late evening arrival peak of flights returning back from Central and Southern Europe.

- **In the general case, regardless of the runway mode of use (segregated, or mixed),** the reduction of capacity due to Wx would cause reactionary delays that would flatten the arrival and departure peaks and push them to the next hour. The Central Office for Delay Analysis (CODA) assessment based on traffic data has confirmed this speculation, based on expert judgement. Given that Helsinki airport has no night curfew, the delayed arrival peak in the evening would simply materialise later. Problem arises when the delayed departure flights risk to overshoot a night curfew at the destination airport. Night curfews are the norm in Europe, rather than an exception.



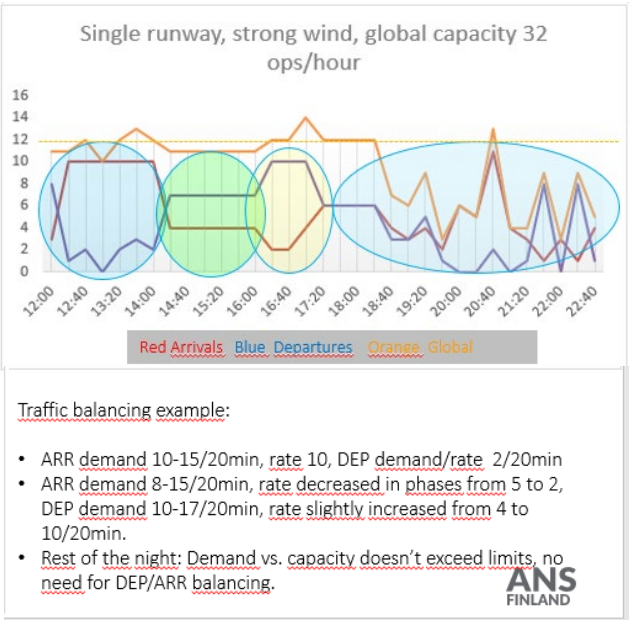


- **In the case of a single runway operations in mixed mode with snow and two snow clearing per hour,** the runway capacity would be significantly reduced, which will cause constant and low arrival and departure rates throughout the whole day, until the Wx situation improves.



- **The case of a single runway operations in mixed mode**, affected by snow, high winds, convective activity, etc. represents the most complex case in terms of balancing the arrival and departure rates.

  The initial stage is to prioritise the arrival peak, making allowances for its longer duration due to the limited runway capacity, or ATFM AD regulation. In the middle stage, nearing the estimated end of the arrival peak, the departure flow will have to be prioritised, while still reserving some spare capacity to cater for the delayed arrivals from the original peak. At the last stage of the balancing exercise, the departure wave would be fully prioritised, because at this time the arrivals of the delayed peak should have landed and completed the turn-around process. And secondly, delaying the departure wave for too long would result in more serious reactionary delay.



- In winter operations when a single runway is in use with two snow removal breaks of 12 min each, the balancing of arrival and departure rates is at premium. **Helsinki has implemented an effective solution using the pre-departure sequencer of the A-CDM platform to account for different ratios between arrivals and departures: 1(A):1(D), 2(A):1(D), or 1(A):2(D).** The ATC SUP playbook table shows the A-CDM parameter settings to manage the departure flow in correlation to the Arrival flow. This is an example how the A-CDM platform could be used as runway capacity management tool for winter operations.